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Integrating enhanced feature fusion with adaptive hidden semi-markov model and deep neural network for robust deepfake detection in Internet of Things-based edge computing environments

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Jawhara Aljabri

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Integrating Enhanced Feature Fusion with Adaptive Hidden Semi-Markov Model and Deep Neural Network for Robust Deepfake Detection in Internet of Things-based Edge Computing Environments

Abstract

The increasing convergence of Internet of Things (IoT) devices and edge computing platforms in current digital networks has presented extraordinary opportunities. However, this surge has presented a massive attack surface susceptible to developing attacks like deepfakes. Deepfakes refer to realistic, but fake images, videos, and sounds made by artificial intelligence (AI) models. Recent developments in deepfake generation have made deepfakes more accurate and simpler to make. Deepfakes have previously been employed for numerous malicious reasons, including the publication of fabricated outcomes in scientific magazines or an attack on the identity tests employed by banks over artificial videos and voices, raising concerns regarding them and their usage. In response to this occurrence, the research group has selected the growth of techniques to distinguish real content from deepfakes. Machine learning (ML) and deep learning (DL) driven systems have attained extremely precise outcomes in the detection of deepfakes. Furthermore, numerous previous studies on perceiving deepfake images or videos were concentrated on DL. This paper presents a Feature Fusion Integrated with Adaptive Hidden Semi-Markov Model for Robust Deepfake Detection (FFAHSMM-RDD) method. The aim is to present a reliable pathway for securing multimedia communication and preventing deepfake-driven threats in real-world applications. Initially, data augmentation, image resizing, and normalization techniques are applied to improve image quality and enhance generalization. For feature representation, a fusion of three models, such as ResNet-50, EfficientNet-B4, and MobileNetV3, is used. Moreover, the adaptive hidden semi-Markov model (AHSMM) is employed to model temporal dependencies and sequence dynamics within the data effectively. In the deepfake classification process, a deep belief network (DBN) model is implemented to classify real and fake content efficiently. The performance validation of the FFAHSMM-RDD methodology demonstrated a superior value of 98.61% under the Celeb DF (v2) dataset

Keywords

Deepfake; Edge Computing; Deep Learning; Internet of Things; Deep Belief Network

1. Introduction

In recent years, owing to advancements in software design and hardware, information and communication technology (ICT) has taken a significant leap, enhancing the efficiency of operations and city lifestyles, leading to the “Smart City”. Different investors are communicated with through the IoT [1]. The percentage of the world population living in cities will reach 70% by the year 2050. The smart city presents a variety of services to the benefit of its citizens [2]. Therefore, people want to be smart to help each smart city service. The maximum potential of smart cities is accessed when a digital ID is delivered to people for accessing all applications or services. Biometrics digital ID is unique and individual-based biometrics are linked to physical or personality traits of a human being [3]. Consequently, it is not required to maintain any private key for protecting the privacy and user identity, as the individual is their private key. Therefore, a digital ID that depends on biometrics could be a smart decision for unlocking the locked door of a smart city. The major difficulties of digital ID are validating an actual individual [4]. Deepfakes, synthetic media established across developed AI techniques, have transformed the visual creation of content by enabling the perfect manipulation of voices, faces, and gestures with remarkable realism [5]. The incorporation of a face recognition approach within organised structures for deepfake creation is an essential improvement, providing ethical protections, authenticity, and improved accuracy. Moreover, the propagation of deepfake content raises important ethical and social problems, involving spreading a rumor, identity theft, and invasion of privacy [6]. There is an urgent need for strong regulatory mechanisms and technical countermeasures to reduce the risks involved with deepfake technology.

Furthermore, current research is concentrated on evolving modern detection and authentication approaches to differentiate between manipulated media and genuine protection against hateful utilisation of deepfake technology [7]. With technological developments, deepfakes become more challenging to differentiate from unique content because of their regular usage of minor alterations that are untraceable to both individuals [8]. ML methods can classify slight variations, which allows precise and effective detection of influenced material, involving deepfake detection. Timely forensic methods concentrate on detecting precise, specific discrepancies, namely pixel-level anomalies, compression artifacts, and unnatural lighting. While these approaches sustain importance, they frequently struggle against extremely advanced manipulations [9]. To address these restrictions, DL methods use information-based patterns, learning to recognize specific discrepancies unnoticeable to the human eye. Various past research on detecting deepfake video or photo was concentrated on

DL approaches, which are divided into 2 detection approaches: Region Convolutional Neural Networks (RCNN)-driven approach and Convolutional Neural Network (CNN)-driven approach. The CNN-driven method takes facial images from video streams and feeds them into the CNN to train and predict to obtain a picture-level outcome [10]. Therefore, in deepfake video streams, some techniques only use spatial data from an individual frame. The RCNN-driven methods need a sequence of video streams to train for producing video-level outcomes. These methods, called CNN, integrate RCNN and Recurrent Neural Networks (RNN). Therefore, the RCNN-driven approach can entirely employ temporal and spatial data in deepfake films [11].

This paper presents a Feature Fusion Integrated with Adaptive Hidden Semi-Markov Model for Robust Deepfake Detection (FFAHSM-M-RDD) method. The key contributions are listed below.

- Initially, the image quality and generalization are improved by applying data augmentation, image resizing, and normalization techniques.
- Furthermore, a fusion of ResNet-50, EfficientNet-B4, and MobileNetV3 is used for feature representation. This extracts rich, multi-scale features for improving detection and computational efficiency.
- Moreover, the adaptive hidden semi-Markov model (AHSM-M) model is employed for modelling temporal dependencies and sequence dynamics, thus enabling effective detection of subtle deepfake artifacts.
- Additionally, the deep belief network (DBN) model is implemented to efficiently classify real and fake content, thus capturing intrinsic associations and deepfake patterns. This also enhances accuracy and robustness compared to softmax or Multilayer Perceptron (MLP) classifiers.
- The experimental validation of the FFAHSM-M-RDD method is performed under the Celeb DF (v2) dataset.

2. Review of Existing DeepFake Detection Approaches

Dagar and Vishwakarma [12] advanced a dual-branch method that combines primarily intended characteristic noise with traditional CNN methods. This approach uses a dual-step strategy, in which the first step combines noise features, and the second step combines RGB characteristics utilizing the hierarchical ConvNext method. Also, this study employs a feature extraction methodology for optimizing and refining the performance qualities. Asaduzzaman

et al. [13] proposed a unique hybrid detection structure that integrates speech and face features with multi-modal biometric analysis and feature-level fusion. This recommended method is outstanding at recognizing slight deepfake changes since it combines Bidirectional Long Short-Term Memory (Bi-LSTM)-driven classifiers for temporal examination and CNNs for feature representation. Moreover, the structure utilizes a multi-task learning method that makes it likely for locating and detecting deepfake artefacts concurrently, improving the presentation and interpretability of the method. Lin et al. [14] presented a Quantum-Trained CNN (QT-CNN) structure intended to improve the deepfake audio identification, utilizing the computing capacity of quantum ML (QML). This method uses a hybrid quantum-traditional method, combining Quantum NNs (QNNs) alongside traditional neural structures for optimizing training effectiveness. This approach integrates a new quantum-to-traditional parameter function that efficiently uses quantum conditions. Yasir and Kim [15] presented an effective and lightweight approach to detect deepfake videos and photos, making it appropriate for devices with restricted computation power. To decrease the computational cost generally associated with the DL method, this method combines an ML classifier with texture analysis and keyframing methods, especially in solving privacy risks produced by artificially created "deepfake" photos on social media settings.

Selvaraj et al. [16] recommended resilient training tactics for amplifying generalization abilities. This method is trained employing purposely crafted examples for a deceiving classification method, thus considerably improving its generalization capability. In terms of this difficulty, a new hybrid adversarial training structure was presented, incorporating Virtual Adversarial Training (VAT) with two-generated blurred adversarial training. This integrated structure boosts the method's strength in deepfakes detection made utilizing new DL methodologies. Zhang et al. [17] suggested a Spectral Transformer with a Pyramid Attention (STPA) method for detecting series variations in manipulated facial photos. Particularly, propose a pyramid attention module that combines either channel or spatial attention methods for prioritizing the face area above the background area. Moreover, a spectral Transformer is used simultaneously for extracting local and global features to enable the fine-grained extraction of the face forgery area. Rafique et al. [18] developed an automated system for detecting deep fake images using Error Level Analysis (ELA), CNN, Support Vector Machines (SVM), and K-Nearest Neighbors (KNN) models to distinguish between real and manipulated images accurately. Kavthekar et al. [19] presented a solution that depends on Masked Image Modelling for tackling the issues of Deepfake Detection. This method

contains 2 methods working simultaneously, such as the Masked Autoencoder and the Multi-scale Vision Transformer, ConvNeXt. The new facial quality detection techniques have evolved to assist in enhancing the information quality by addressing the difficulty of misrepresented facial information. Deepfake videos are utilized for different hateful reasons. Table 1 signifies an overview of edge computing-based deepfake detection.

Table 1 Comparative study of deepfake detection's related work

Authors	Goals	Technique	Datasets	Outcomes
Dagar and Vishwakarma [12]	To suggest a dual-branch technique that integrates manually intended feature noise with traditional models. Additionally, it employs loss of edge supervision by taking boundary manipulation data and inducing a precise localization at the edges.	CNN	Shallowfakes and Faceforensics++ Datasets	AUC score of 99%.
Asaduzzaman et al. [13]	To project a unique hybrid recognition method that integrates speech features with fusion and multi-modal biometric recognition. The recommended method is excellent at recognizing subtle modifications in deepfakes.	CNN and Bi-LSTM	DeepfakeTIMIT and AVSpeech Datasets	Accuracy of 98.7%.
Lin et al. [14]	Intended to improve the recognition of deepfake audio, utilizing the computational power and integrating an innovative quantum-to-classical parameter map that effectively employs quantum states to increase powers.	QML and QNN	DEEP-VOICE Dataset	Achieving parameter reduction up to 70%.
Yasir and Kim [15]	To project a lightweight and effective approach for identifying deepfake videos and images makes it suited for gadgets with confined computational resources.	DL	FaceForensics+ and Celeb-DF(v2) Datasets	Accuracy of 92% and 96%.
Selvaraj et al. [16]	To develop robust training tactics by increasing the ability of generalization. In	VAT	FaceForensics+ Dataset	Precision 86.7%.

adversarial training, purposely crafted instances are used to deceive classification models, thus substantially improving their generalizability.

Zhang et al. [17]	To identify sequence permutations in a manipulated facial image. Particularly, develop a pyramid attention module, which combines either spatial or channel attention mechanisms by prioritizing the face region over the background.	ResNet50 and Transformer	Seq-Deepfake Dataset	Accuracy of 72.30% and 68.71%.
Rafique et al. [18]	To explore a trustworthy approach for recognizing deepfake images, multiple investigators are working tirelessly to recognize the content of deep fakes utilizing various approaches.	CNN, SVM, and KNN	DF Dataset	Accuracy of 89.5%.
Kavthekar et al. [19]	To present a solution depends on a masked image model and to address the issue of identification. An innovative recognition model is advanced that aids in increasing the quality of data by overcoming the task of misrepresented facial data.	Auto Encoders and Vision Transformers	DFDCp Face Forensics++ Datasets	Accuracy 80.22% and 98.1%.

3. Proposed Methodological Framework

In this manuscript, an FFAHSMM-RDD model is presented in IoT and Edge Computing Applications. The FFAHSMM-RDD method is created to provide a reliable path to safeguard multimedia communication and prevent deepfake-based threats in real-world applications. To perform that, the FFAHSMM-RDD method encompasses the following processes: Input image processing, feature representation models, sequence modelling using the AHSM method, and classification via the DBN technique. Fig. 1 signifies the general process of the FFAHSMM-RDD method.

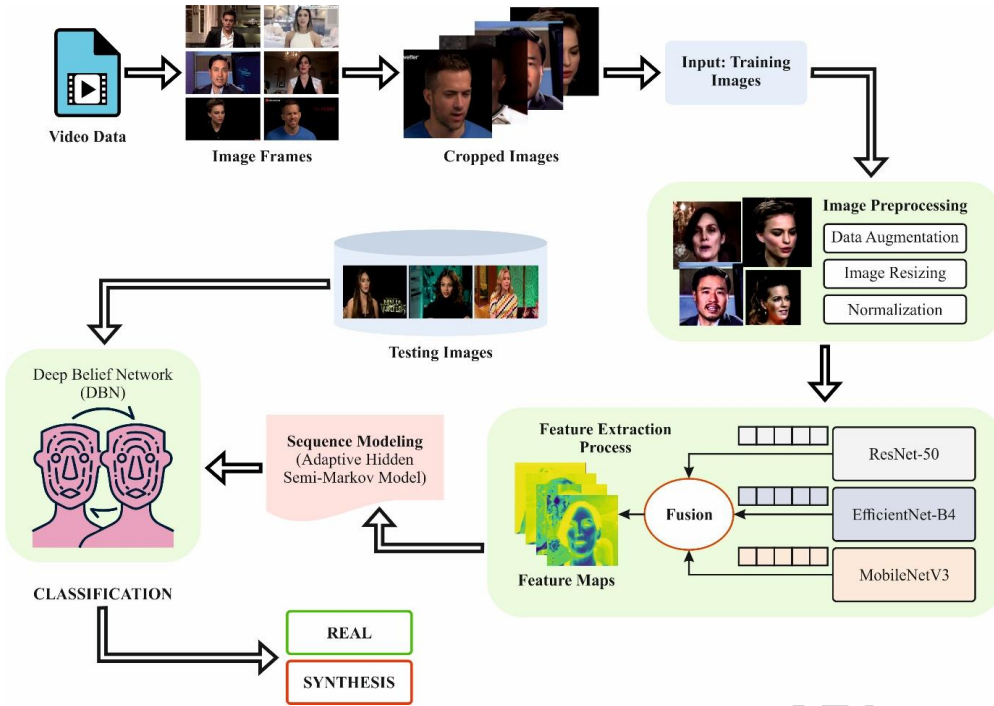


Fig. 1 General process of the FFAHSMM-RDD approach

3.1. Image Pre-processing Methods

Primarily, the collected image data is served into numerous pre-processing steps like data augmentation, image resizing, and normalization [20]. These steps confirm that the data is reliable, clean, and stable for effective learning. These steps are significant to heighten the generality and the accuracy of the model. The brief description of these steps is given below.

Data Augmentation

To increase the assortment of databases and to decrease the problem of overfitting, the set of augmentation models is used, like random rotation, rescaling, zooming, horizontal flipping, shifting of height and width, and alteration of brightness. The model of rescaling is employed to normalize the pixel intensity; the random rotation model will somewhat alternate images to aid the model learn from dissimilar viewpoints. Further, horizontal flipping, vertical flipping, and both are employed to flip an image to simulate manifold directions, which is signified in Eqs. (1)-(3) correspondingly.

$$x', y' = W - 1 - x, y \quad (1)$$

$$x', y' = x, H - 1 - y \quad (2)$$

$$x', y' = W - 1 - x, H - 1 - y \quad (3)$$

Here, x and y denote the location of a pixel, W indicates the width, H is the height of an image, and x' and y' are the new locations of a pixel. Lastly, height and width shifting moves image content in vertical or horizontal direction to pretend spatial variation, and brightness adjustment adjusts image intensity to make the model strong to variances of brightness.

Image Resizing

The raw data of every image is handled by the image resizing mode, which resizes every image to a fixed resolution of 224x224 pixels. The resizing describes the size of the target. In the context of the developed method, the target size is 224 pixels in width and height. Further, the original image's pixels are mapped to the target grid, where the resizing technique computes every novel pixel in a 224x224 grid, and each must get its value from the original image. If an original image is greater than, it downscales the content, and if the original image is smaller, it executes upscaling the content. Therefore, step 3 contains interpolation that contains a weighted average of the 4 nearest pixels from an original image, both vertically and horizontally. Afterward, a novel image of 224x224 pixels is built. This resolution model certifies that every image of input has equivalent sizes, which approves smooth TL without altering the structure of the model.

Normalization

Further, the resized data is served into a normalization model to alter the values of pixel intensity in an image within the range of $[0,1]$ or $[-1,1]$, respectively. The proposed model employed a mean standard normalization model, which measures the values of pixels utilizing a mean and standard deviation. Eq. (4) provides a mathematical calculation.

$$X_{normalized} = \frac{X - \mu}{\sigma} \quad (4)$$

Here, μ signifies mean and σ denotes standard deviation. These normalized values certify that an input image distribution will equal the pre-trained method employed in the proposed structure.

3.2. Fusion of Feature Extraction Techniques

For the feature extraction, a fusion of three methods, including ResNet-50, EfficientNet-B4, and MobileNetV3, is utilized for capturing diverse and discriminative features. This fusion presents robust and effective feature extraction. The features are fused by using a concatenation technique, and also, the methods utilize ImageNet pretrained weights with partial fine-tuning (final layers unfrozen) to adapt to deepfake-specific artifacts while averting overfitting.

3.2.1. ResNet-50 Architecture

The ResNet-50 method is a deep convolutional neural network (DCNN), which consists of 50 layers, intended to tackle the vanishing gradient issue in deep networks [21]. Its structure is constructed on the notion of residual learning, where shortcut connections go around one or many layers. These connections allow the system to learn residual mappings instead of an original feature mapping, permitting deeper layers to concentrate on more intricate features without losing gradient data. The ResNet-50 method is a CNN, which was initially pre-trained on the ImageNet database. The structure of the ResNet-50 model holds 50 layers and is generally divided into 5 sections. The first part contains an input pre-processing layer, which consists of a convolutional layer. The remaining 4 parts contain Bottleneck Building Blocks. Every Bottleneck Building Block comprises numerous normalization layers, ReLU activation functions, convolutional layers, and a shortcut connection. The ResNet-50 technique has 49 convolutional layers and one fully connected (FC) layer in total. The design of Bottleneck Building Blocks, which creates ResNet-50, is stated by $y = F(x) + x$. In the given calculation, $F(x)$ is denoted as the residual function, whereas x and y signify the residual function's input and output parameters. The structure of ResNet uses two kinds of building blocks, regular and modified. The regular building blocks are employed in ResNet-34 and hold dual layers of 3×3 convolutional layers.

The bottleneck building blocks are a specific alteration employed in deeper ResNet structures like ResNet-50, ResNet-101, and ResNet-152. These blocks contain a series of 3 layers, such as 1×1 , 3×3 , and 1×1 convolutional layers. The dual 1×1 layers aid in decreasing and then enlarging the sizes, thus narrowing the flow's dimensions. This makes a bottleneck effect, which provides a Bottleneck Building Block. This modification permits a deeper network to be trained without considerably enlarging the time of training. The feature mapping of ResNet-50 exposes intricate and expanded activation patterns in the deeper layer owing to its deep structure and residual connections.

3.2.2. EfficientNet-B4 Model

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EfficientNet includes a sequence of CNN structures, which attain advanced accuracy with significantly enhanced computational efficacy over preceding methods [22]. The main invention of EfficientNet rests in its compound scaling model that evenly measures the network's width, depth, and input resolution utilizing a group of fixed scaling coefficients. This evades the physical and frequently sub-optimal scaling of individual sizes observed in previous structures. The base method, EfficientNet-B0, is constructed utilizing the Mobile Inverted Bottleneck Convolution (MBConv) blocks presented in MobileNetV2, which unite depth-wise separable convolution with squeeze- and-excitation (SE) optimization to enrich channel-wise feature recalibration. Also, it uses Swish activation for enhanced non-linear expressiveness. Subsequent methods in the EfficientNet family (B1-B7) originate from analytically scaling up the base structure with a compound coefficient, upholding the balance between efficiency and accuracy. The compound scaling model at EfficientNet measures the network's width, depth, and input resolution utilizing a fixed coefficient set. Instead of randomly altering 1D, this model seeks to enhance every dimension, governed by the below-mentioned formulations:

$$\text{depth} = \alpha^\phi, \quad \text{width} = \beta^\phi, \quad \text{resolution} = \gamma^\phi \quad (5)$$

Subject to constraint:

$$\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \text{ with } \alpha \geq 1, \beta \geq 1, \gamma \geq 1 \quad (6)$$

Here, ϕ denotes a compound coefficient.

$$\alpha = 1.2, \quad \beta = 1.1, \quad \gamma = 1.15 \quad (7)$$

EfficientNet-B4 corresponds to the compound coefficient of $\phi = 4$, which results in the below-mentioned scaled factors:

$$\begin{aligned} \text{depth scale} &\approx 1.2^4 = 2.07 \\ \text{width scale} &\approx 1.1^4 = 1.46 \\ \text{resolution scale} &\approx 1.15^4 = 1.75 \end{aligned} \quad (8)$$

Therefore, EfficientNet-B4 originates by evenly scaling up EfficientNet-B0 by these factors, which leads to a method with an input resolution of 380×380 pixels and nearly 19 million parameters. This principled scaling upholds the balance between model capability and the

cost of computation. EfficientNet-B4 holds initial convolutional layers tracked by a sequence of MBConv blocks with fluctuating kernel sizes (3×3 and 5×5) and expansion factors. Every group of blocks is marked with the number of layers, gradually constructing a hierarchical feature representation down to the last feature mapping. Fig. 2 describes the framework of the EfficientNet-B4 method.

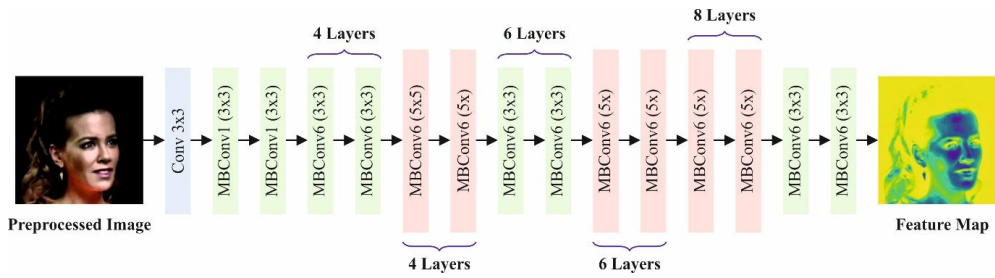


Fig. 2 Framework of the EfficientNet-B4 method

3.2.3. MobileNetV3 Method

MobileNet-V3 is an effective creation of a DL method for edge computing. It holds dual releases, such as larger and smaller [23]. The larger version owns a somewhat higher number of parameters and is more challenging, which permits it to acquire several intricate patterns and high-scale abstract features. Therefore, the larger method was selected as the basis encoding in this work. Constructing upon the inventions of its predecessors, MobileNet-V3 was more sophisticated over the combination of dual superior models: MNas Net, which enhances configuration assortment, and NetAdapt model, intended at finding and perfecting network structure. In a first phase, MobileNet-V3 executes an innovative activation function recognized as hard swish (*h-swish*) that improves speed of operation and decreases computational cost when compared to an improved sigmoid function. The main building block is named as an inverted residual block that unites a depth-wise separable convolutional block with an attention block of a squeeze-and-excitation (SE) network. This intention appeals to stimulation from a layer of bottleneck, using an inverted residual connection for connecting input as well as output features in similar networks, thus enhancing feature depiction while diminishing the usage of memory. Compared to a conventional convolution block, depth-wise separable convolution uses a depth-wise kernel for every network, followed by a 1×1 point-wise convolution kernel and a layer of batch normalization. The SE block improves the model's ability to concentrate on major feature networks, thus reinforcing its representative power. This procedure includes calculating the mean of an input feature

over a layer of average pooling, using h -swish or ReLU activation functions for originating feature weights, and then multiplying it by a matrix of original features to produce a weighted output feature. This model considerably supports MobileNet-V3's learning efficiency by merging feature networks.

In the last phase, an average pooling layer is innovative, and a convolutional layer alters the number of feature networks to enlarge into a high-dimensional space. This increases the pace of computation and performance without compromising accuracy. With its lightweight structure and extraordinary performance, MobileNet-V3 has become an effective backbone in applications of industrial and academic use.

3.3. Adaptive Hidden Semi-Markov Model

Furthermore, the AHSMM method is used for effectively modeling temporal dependencies and sequence dynamics in the data [24]. This method efficiently captures state durations and also assists in detecting consistent deepfake artifacts. The consecutive frames are used for forming video sequences. Also, a CNN feature vector is employed for the representation of each frame. These vectors also act as constant observation input. Furthermore, this model is excellent in modelling duration-specific patterns. Moreover, the hidden states link the temporal modelling to deepfake behaviour.

The HSMM is an addition of HMM with the inclusion of a variable sojourn time. For each fake state, several observations are produced based on the sojourn time of the fake state. Here, the sojourn time of each fake condition is set by a Weibull distribution. The parameters of HSMM are defined below:

- N : number of fake states. The set of fake states is represented as $S = \{S_1, S_2, \dots, S_N\}$. Every fake state signifies a frame consistency level.
- M : number of different observation symbols. The observation procedure is demonstrated with a Gaussian distribution; hence, the space contains every real number.
- λ : function of transition rate. It defines the frame consistency procedure, which follows a rate function, represented by λ . Here, the λ function relates to the Weibull distribution, which is regularly employed in reliability theory.

To train HSMM, the Expectation-Maximization algorithm is employed. Some assumptions are prepared to use this method for prognostics.

- Initial state: The model begins in the initial fake state. So, the engineering method is expected to continue as good as new.
- Transitions: only left-to-right transitions are permitted. This means that the method can only transfer to a neighboring state or sojourn in the present state. So, it is expected that the engineering method does not recover health and that it has to go over every state before reaching failure.
- Final state: the final state of HSMM is visible, and it signifies failure. Here, only one value of observation is released.

As previously stated, the model AHSMM is an adaptive addition of HSMM. Throughout the testing stage, the adaptation is generated once a changeover from the fake state occurs, and the transition from the fake state S_i to the fake state S_{i+1} happens. Then, the sojourn time of the fake state S_i is T_i and is equated to an expected value from the Weibull distribution for that fake state, E_i . The Weibull distribution is expressed in Eq. (9), in which α is a shape parameter and β denotes a scale parameter. The expected value is formulated in Eq. (10).

$$f(x) = \begin{cases} \frac{\alpha}{\beta} \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-\left(\frac{x}{\beta}\right)^\alpha}, & x \geq 0, \\ 0, & x < 0. \end{cases} \quad (9)$$

$$\mathbb{E}[X] = \beta \Gamma\left(1 + \frac{1}{\alpha}\right) \quad (10)$$

A ratio between T_i and E_i of the present state, S_i , is intended and employed as a resampling factor (R_f) for the sojourn time of the following fake state, S_{i+1} . To resample the sojourn time of S_{i+1} , the scale parameter is modified since it can move the Weibull distribution to have smaller and extended sojourn times. The adapted scale parameter for S_{i+1} is signified as β_{i+1}^* and explained in Eq. (11). Therefore, the Weibull distribution for S_{i+1} is well-defined by parameters $(\alpha_{i+1}, \beta_{i+1}^*)$, as the shape parameter α is not modified and stays similar to the one calculated throughout training.

$$\beta_{i+1}^* = \frac{E_{i+1}[R_f]}{\Gamma\left(1 + \frac{1}{\alpha_{i+1}}\right)} \quad (11)$$

Eq. (11) moves the Weibull so that its predictable sojourn equals the experimental behavior detected in deployment, permitting the method to shorten or lengthen forecast segment periods after unexpected events adaptively. This adaptation enhances detection robustness in an edge environment where operations can differ from training.

3.4. DBN-based Classification

In the deepfake classification process, a DBN is employed to classify real and fake content [25] effectively. This method is chosen for its effectiveness in learning hierarchical, generative data transformation from the fused CNN and AHSMM models, thus capturing intrinsic associations in deepfake patterns. This model is also considered for its efficiency over conventional classifiers. Furthermore, this model is efficient for computation, and unlike transformers, they are excellent in modelling high-level abstractions. This mitigates overfitting and stabilizes learning. The DBN model also enhances accuracy and robustness against subtle changes in videos, while also extracting probabilistic latent features.

DBN is formed by many layers of stacked Restricted Boltzmann Machines (RBM), proficient in removing deeper features from the original data. Pre-training comprises unsupervised learning; here, the procedure initiates the input data passed into the primary RBM. The output from the primary RBM turns the input for the second RBM, and this layer-wise training remains until each layer is completely trained. After pre-training, the parameters of every layer, like biases and weights, were employed by the primary values for the parameters of the network. The training model initiates with a low-level RBM, which acquires the inputs of DBN and develops higher levels by the hierarchy till the final RBM is trained at the topmost layers that comprises the output of DBN. In this RBM, the middle and visible layer l_h and l_v comprise random dual parameters in the middle and upper layers, respectively. Furthermore, the indirect weight connects the visible layer and middle layer (w), which were designed to create a shared distribution function.

$$\rho(l_v, l_h) = \frac{1}{F_p} e^{-E(l_v, l_h)} \quad (12)$$

While F_p denotes the partition function, computed by adding each probable pair of middle and observable units.

$$F_p = \sum_{l_v, l_h} e^{-E(l_v, l_h)} \quad (13)$$

The total energy of the combined state of the hidden unit and visible unit is presented.

$$E(l_v, l_h) = - \sum_{i=1} a_i l_{vi} - \sum_{j=1} b_j l_{hj} - \sum_{i,j} l_{vi} w_{ij} l_{hj} \quad (14)$$

Here, l_{hj} and l_{vi} refer to the binary state and position of hidden nodes j and observable nodes i , respectively, and w_{ij} depicts the weighted connection of the observable layer and the middle layer, respectively. Furthermore, a_i and b_j signify the biases for observable and middle nodes, respectively.

The RBM's weight is further refined to employ the succeeding equation:

$$\Delta w_{ij} = E_t(l_{vi}l_{hj}) - E_m(l_{vi}l_{hj}) \quad (15)$$

Now, $E_t(\cdot)$ and $E_m(\cdot)$ depict the expectation for learning data and models, correspondingly. Whereas DBN is effective for multiple tasks, accomplishing an optimum solution depends upon having a suitably designed framework. Algorithm 1 describes the DBN model.

Algorithm 1: DBN method

Input Preparation:

- Use input data.

2. Layer-wise Pre-training (Unsupervised):

- Send input to the initial RBM and train it for learning hidden feature representations.
- Give the output of RBM to the subsequent RBM.
- Repeat layer-wise until all RBMs are given training, thus forming the hierarchical DBN.

3. Parameter Initialization:

- Initialize the corresponding DBN layers by using the trained weights and biases from every RBM.

4. Fine-tuning (Supervised):

- Form the full DBN for stacking all RBMs.
- Train the DBN end-to-end on labeled data for optimizing the process.

5. Feature Hierarchy:

- Capture simple features by utilizing lower RBMs.

6. Output Prediction:

- Obtain the output as real or deepfake in the final layer.

3.5. Evaluation Metrics

For evaluating the proposed method's performance, recall, accuracy, specificity, and AUROC are used as the main measures. To calculate the accuracy, the number of samples properly

categorized and divided by the total number of instances. It is computed in the mathematical formulation as exposed in Eq. (16). Here, tn , tp , fn , and fp imply true negative, true positive, false negative, and false positive, respectively. Thus, having a higher score in both recall and sensitivity is vital. Precision is essential for defining how many instances predicted to be positive are really positive. Precision was computed by separating the number of tp by the summation of fp and tp as shown in Eq. (17).

$$Acc = \frac{tp + tn}{tn + tp + fn + fp} \quad (16)$$

$$Pre = \frac{tp}{tp + fp} \quad (17)$$

Besides, the classifier utilizes recall or sensitivity to compute the proportion of real positive cases, which are accurately recognized as true positives. Eq. (18) portrays the numerical model for recall. The F1 score was defined by utilizing their precision and recall as described in Eq. (19). Furthermore, the confusion matrix aids in assessing how fine a classification model achieves by attaining incorrect as well as correct predictions it creates. It has dual axes, such as true and predicted labels. Likewise, the ROC curve is computed by equating the false positive rate with the true negative rate. The AUC assesses the ROC curve efficiency in categorizing an input. If the value of AUC increases, the performance of the model's classification increases. The AUC was defined for every model, and lastly, each model is equated to define which one beats the other.

$$Rec = \frac{tp}{tp + fn} \quad (18)$$

$$F1 - score = \frac{2tp}{2tp + fp + fn} \quad (19)$$

4. Experimental Setup

In this section, the experimental validation of the FFAHSMM-RDD model is verified under the Celeb DF (v2) dataset [26]. The model is simulated using Python 3.6.5 on a PC with an i5-8600k, 250GB SSD, GeForce 1050Ti 4GB, 16GB RAM, and 1TB HDD. Parameters include a learning rate of 0.01, ReLU activation, 50 epochs, 0.5 dropout, and a batch size of 5. The training and testing analysis ensure the prevention of data leakage and overfitting. Also, temporal dynamics are effectively captured from each video at regular intervals. Furthermore, techniques such as data augmentation, regularization, and partial fine-tuning are

used for reducing overfitting, thus retaining general features. Additionally, data augmentation and weighted loss functions are employed for handling the data imbalance. This dataset comprises real and DeepFake-synthesized videos having the same visual quality as those circulated online. This dataset contains 12 real and synthetic videos. The training has 4004 real and 4027 synthesis images. And the testing has 1716 real and 1726 synthetic images. The complete details of this dataset are shown in Table 2. Fig. 3 exhibits a sample of real and synthesized images. Fig. 4 illustrates the original images, pre-processed images, and feature maps of set 1, and Fig. 5 portrays the original images, pre-processed images, and feature maps of set 2.

Table 2 Details of the dataset

No. of Video		No. of Images		Testing	
Real	Synthesis	Training		Real	Synthesis
12	12	Real 4004	Synthesis 4027	1716	1726

(a)
(b)

Fig. 3 Sample of (a) Real and (b) DeepFake images



Fig. 4 Original images, pre-processed images, and feature maps on set 1

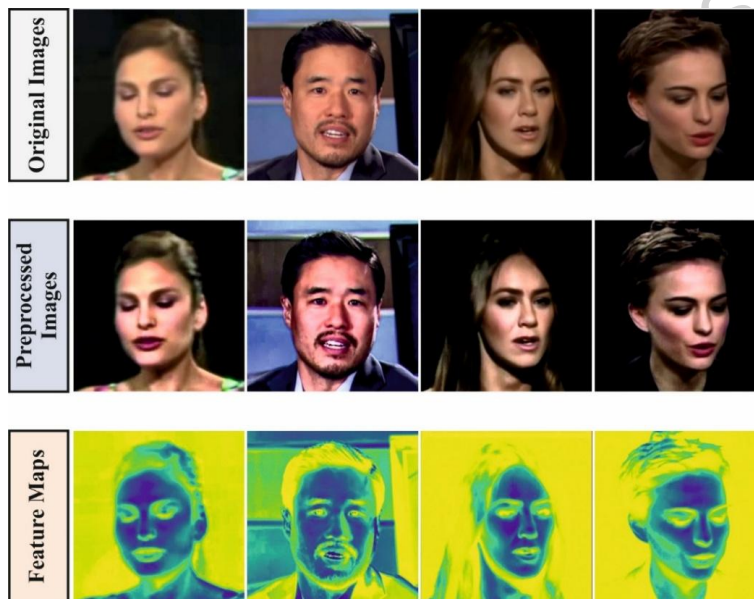


Fig. 5 Original images, pre-processed images, and feature maps on set 2

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4.1. Classification Results of Proposed Model

Fig. 6 displays the classifier outcome of the ResNet-50 model. Figs. 6a and 6b show the confusion matrices at 70:30. The figure implies that the ResNet-50 model has recognized two

classes precisely. Fig. 6c demonstrates the training and validation accuracy over 100 epochs. The figure shows that the ResNet-50 model achieves progressively higher accuracy values over rising epochs. The results specify that the ResNet-50 technique accomplishes close training and validation loss values. Fig. 6d exhibits the training and validation loss over 100 epochs. Fig. 6e illustrates the PR examination, denoting maximal performance on each class. Fig. 6f exemplifies the ROC investigation, exhibiting efficacious outcomes with higher ROC values for diverse classes. At Last, Figs. 6g and 6h present the classification outcome at 70:30. The results specify that the ResNet-50 technique has effectively detected the real and synthesized samples under all classes.

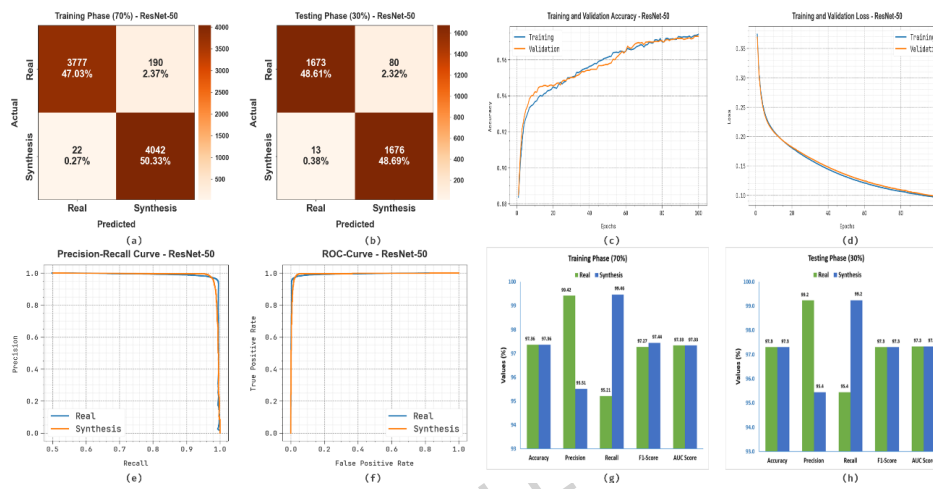


Fig. 6 ResNet-50 model (a, b) Confusion matrices, (c, d) Accuracy and Loss curves, (e, f) PR and ROC curves, and (g, h) Training and Testing phases of 70:30

Table 3 and Fig. 7 represent a deepfake detection on the ResNet-50 method. On 70% TRPHE, it obtains average $accuracy$, $precis$, $recall$, $F1_{score}$, and AUC_{score} of 97.36%, 97.47%, 97.33%, 97.36%, and 97.33%, respectively. Similarly, at 30% TSPHE, it attains average $accuracy$, $precis$, $recall$, $F1_{score}$, and AUC_{score} of 97.30%, 97.34%, 97.33%, 97.30%, and 97.33%, respectively.

Table 3 Deepfake detection of the ResNet-50 method

Classes	$Accuracy$	$Precis$	$Recall$	$F1_{score}$	AUC_{score}
TRPHE (70%)					
Real	97.36	99.42	95.21	97.27	97.33
Synthesis	97.36	95.51	99.46	97.44	97.33
Average	97.36	97.47	97.33	97.36	97.33
TSPHE (30%)					

Real	97.30	99.23	95.44	97.30	97.33
Synthesis	97.30	95.44	99.23	97.30	97.33
Average	97.30	97.34	97.33	97.30	97.33

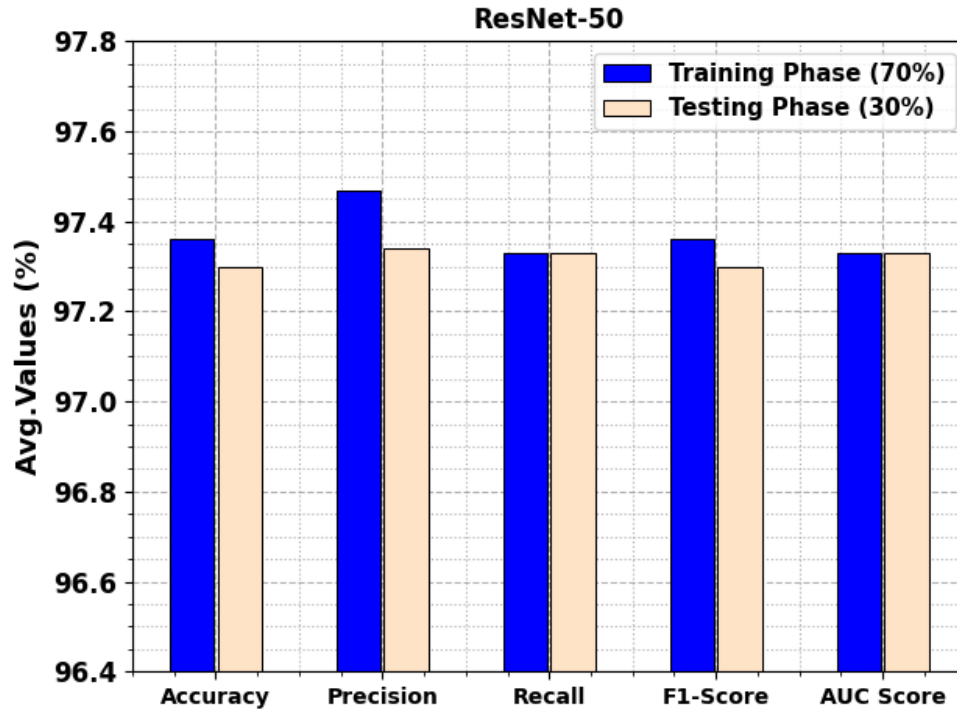


Fig. 7 Average values of the ResNet-50 method under 70:30

Fig. 8 points out the classifier outcome of the EfficientNet-B4 method. Figs. 8a and 8b portray the confusion matrices at 70:30. The figure denotes that the EfficientNet-B4 model accurately identified dual classes. Fig. 8c describes the training and validation accuracy over 100 epochs. The figure shows that the EfficientNet-B4 technique gets higher accuracy values over growing epochs. Fig. 8d displays the training and validation loss over 100 epochs. The outcomes imply that the EfficientNet-B4 technique attains close training and validation loss values. Fig. 8e exhibits the PR inspection, denoting maximal performance on every class. Fig. 8f demonstrates the ROC study, displaying efficacious outcomes with the highest ROC values for various classes. Finally, Figs. 8g and 8h embody the training and testing phases at 70:30. The results point out that the EfficientNet-B4 technique has effective detection of the real and synthesis samples on all classes.

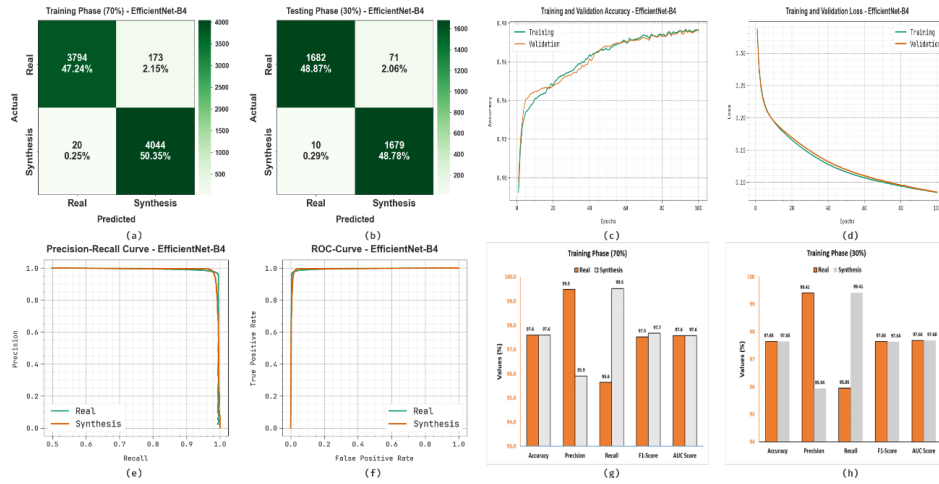


Fig. 8 Results of EfficientNet-B4 method (a, b) Confusion matrices, (c, d) Accuracy and Loss curves, (e, f) PR and ROC curves, and (g, h) Training and Testing phases of 70:30

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Table 4 and Fig. 9 represent a deepfake detection of the EfficientNet-B4 method. At 70% TRPHE, it obtains average $accu_r$, $preci_n$, $recal_l$, $F1_{score}$, and AUC_{score} of 97.60%, 97.69%, 97.57%, 97.59%, and 97.57%, respectively. Likewise, on 30% TSPHE, it achieves average $accu_r$, $preci_n$, $recal_l$, $F1_{score}$, and AUC_{score} of 97.65%, 97.68%, 97.68%, 97.65%, and 97.68%, respectively.

Table 4 Deepfake detection of the EfficientNet-B4 approach

Classes	$Accu_r$	$Preci_n$	$Recal_l$	$F1_{score}$	AUC_{score}
TRPHE (70%)					
Real	97.60	99.48	95.64	97.52	97.57
Synthesis	97.60	95.90	99.51	97.67	97.57
Average	97.60	97.69	97.57	97.59	97.57
TSPHE (30%)					
Real	97.65	99.41	95.95	97.65	97.68
Synthesis	97.65	95.94	99.41	97.64	97.68
Average	97.65	97.68	97.68	97.65	97.68

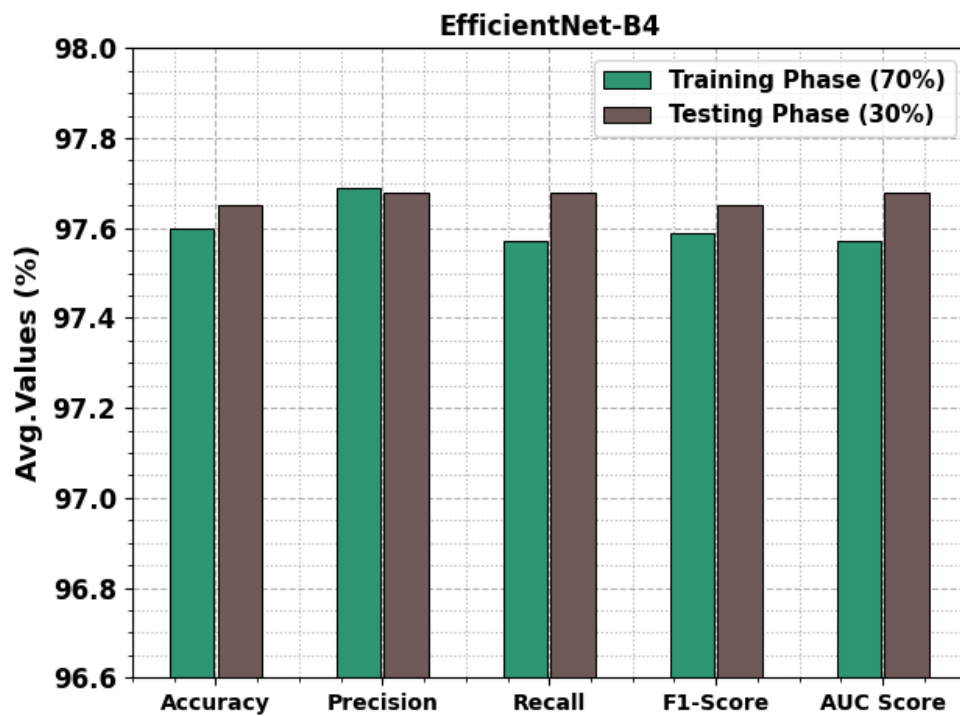


Fig. 9 Average values on the EfficientNet-B4 approach under 70:30

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Fig. 10 portrays the classifier outcome of the MobileNetV3 approach. Figs. 10a and 10b describe the confusion matrices at 70:30. The figure implies that the MobileNetV3 model has identified all 2 class labels correctly. Fig. 10c illustrates the training and validation accuracy over 100 epochs. The figure reports that the MobileNetV3 model achieves greater accuracy values over extended epochs. Fig. 10d displays the training and validation loss over 100 epochs. The results denote that the MobileNetV3 model attains closer training and validation loss values. Fig. 10e reveals the PR investigation, denoting maximal performance on each class. Fig. 10f proves the ROC analysis, indicating capable outcomes with the highest ROC values for separate classes. Figs. 10g and 10h signify the training and testing phases at 70:30. The outcomes suggest that the MobileNetV3 model has effectively recognized the real and synthesized samples under all classes.

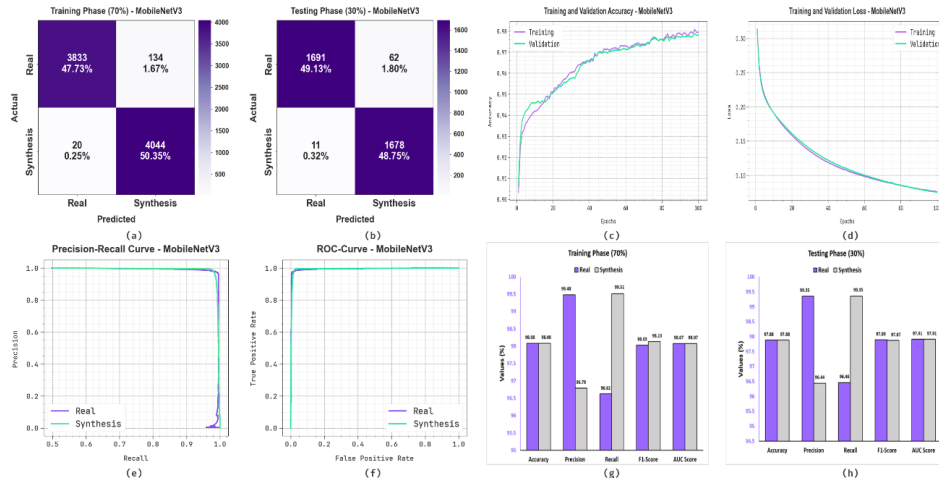


Fig. 10 Results of MobileNetV3 method (a, b) Confusion matrices, (c, d) Accuracy and Loss curves, (e, f) PR and ROC curves, and (g, h) Training and Testing phases of 70:30

Table 5 and Fig. 11 present a deepfake detection of the MobileNetV3 method. On 70% TRPHE, it gets average $accu_r$, $preci_n$, $recal_l$, $F1_{score}$, and AUC_{score} of 98.08%, 98.14%, 98.07%, 98.08%, and 98.07%, respectively. Also, at 30% TSPHE, it gets average $accu_r$, $preci_n$, $recal_l$, $F1_{score}$, and AUC_{score} of 97.88%, 97.90%, 97.91%, 97.88%, and 97.91%, respectively.

Table 5 Deepfake detection of the MobileNetV3 method

Classes	$Accu_r$	$Prece_n$	$Recal_l$	$F1_{score}$	AUC_{score}
TRPHE (70%)					
Real	98.08	99.48	96.62	98.03	98.07
Synthesis	98.08	96.79	99.51	98.13	98.07
Average	98.08	98.14	98.07	98.08	98.07
TSPHE (30%)					
Real	97.88	99.35	96.46	97.89	97.91
Synthesis	97.88	96.44	99.35	97.87	97.91
Average	97.88	97.90	97.91	97.88	97.91

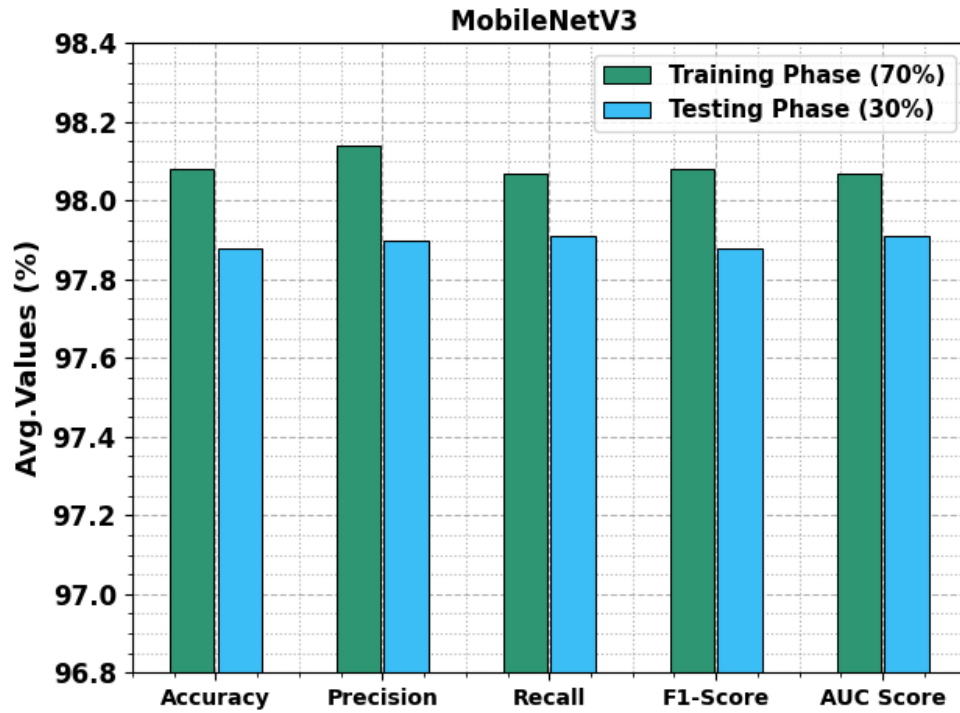


Fig. 11 Average values on the MobileNetV3 method under 70:30

Fig. 12 exemplifies the classifier outcome of the Fusion model. Figs. 12a and 12b depict the confusion matrices under 70:30. The figure denotes that the Fusion model has classified all 2 class labels accurately. Fig. 12c shows the training and validation accuracy over 100 epochs. The figure reports that the Fusion model obtains rising accuracy values over extending epochs. Fig. 12d illustrates the training and validation loss over 100 epochs. The results specify that the Fusion model achieves close values of training and validation loss. Fig. 12e reveals the PR examination, denoting maximal performance on each class. Fig. 12f portrays the ROC investigation, determining proficient outcomes with superior ROC values for individual classes. Finally, Figs. 12g and 12h denote the training and testing phases at 70:30. The results denote that the Fusion model has effective identification of the real and synthesis samples under all classes.

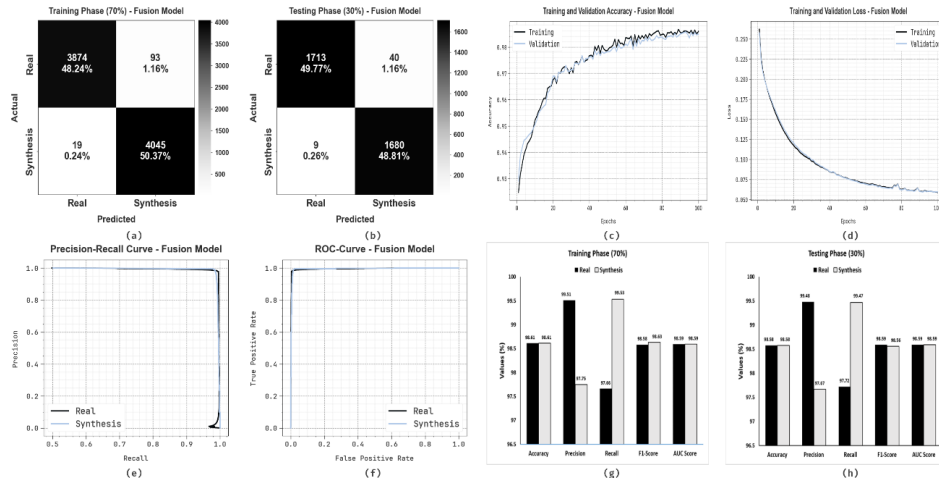


Fig. 12 Results of Fusion model (a, b) Confusion matrices, (c, d) Accuracy and Loss curves,

(e, f) PR and ROC curves, and (g, h) Training and Testing phases of 70:30

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Table 6 and Fig. 13 display a deepfake detection of the Fusion model. Under 70% TRPHE, the fusion model accomplishes average $accuracy$, $precision$, $recall$, $F1_{score}$, and AUC_{score} of 98.61%, 98.63%, 98.59%, 98.60%, and 98.59%, respectively. Further, at 30% TSPHE, the fusion model obtains average $accuracy$, $precision$, $recall$, $F1_{score}$, and AUC_{score} of 98.58%, 98.58%, 98.59%, 98.58%, and 98.59%, respectively.

Table 6 Deepfake detection of Fusion model

Classes	$Accuracy$	$Precision$	$Recall$	$F1_{score}$	AUC_{score}
TRPHE (70%)					
Real	98.61	99.51	97.66	98.58	98.59
Synthesis	98.61	97.75	99.53	98.63	98.59
Average	98.61	98.63	98.59	98.60	98.59
TSPHE (30%)					
Real	98.58	99.48	97.72	98.59	98.59
Synthesis	98.58	97.67	99.47	98.56	98.59
Average	98.58	98.58	98.59	98.58	98.59

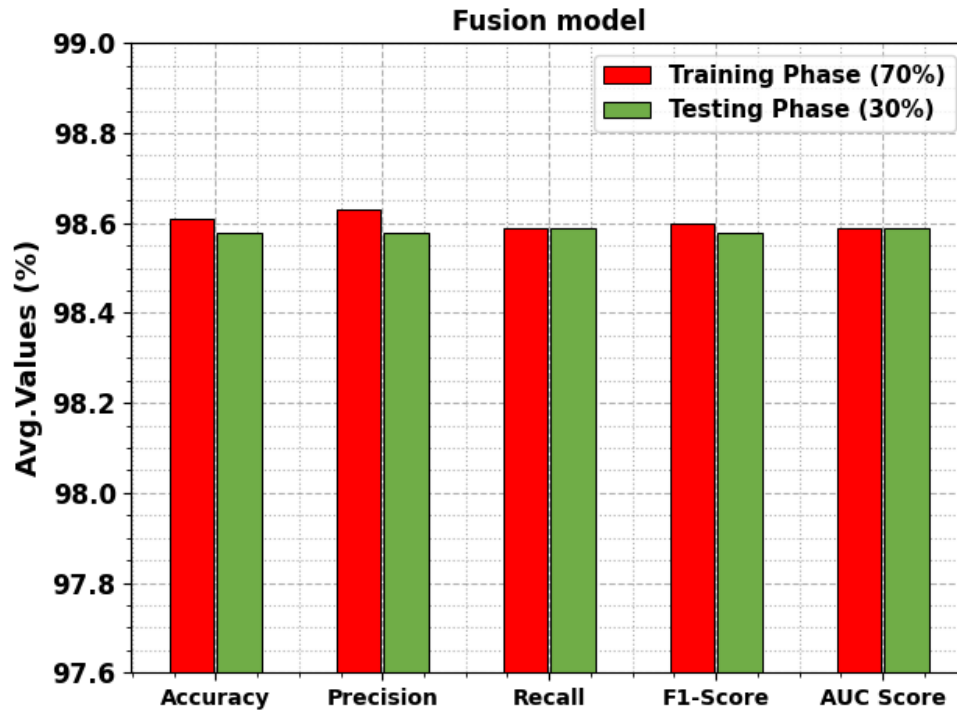


Fig. 13 Average values of the Fusion model under 70:30

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4.2. Comparative Result Analysis with Existing Studies

Table 7 exemplifies the comparative investigation of the FFAHSMM-RDD model with existing models under training and testing [27-33]. Fig. 14 presents a comparison of the outcome of the FFAHSMM-RDD model with the existing model under training. The table values specify that the FFAHSMM-RDD methodology achieved maximum performance with $accuracy$, $prec_i_n$, and AUC_{Score} of 98.61%, 98.63%, and 98.59%, respectively.

Table 7 Comparative analysis of FFAHSMM-RDD model with existing approaches

Model	Training			Testing		
	$Accur_y$	$Preci_n$	AUC_{Score}	$Accur_y$	$Preci_n$	AUC_{Score}
CNN+ SVM [27]	88.10	90.27	95.76	95.95	97.89	89.14
Custom CNN model [28]	97.24	90.94	91.06	89.02	93.60	91.36
CEViT-VDD [30]	87.06	93.99	95.79	90.06	94.72	98.38
FFDL-DVD [31]	84.56	92.85	97.79	91.28	95.76	90.12
DVD-CViT [32]	90.35	97.74	93.23	94.24	98.05	94.93
ViXNet-DVIFD [29]	94.40	96.52	95.52	93.35	96.09	91.80
DD-DFSML [33]	96.33	91.84	96.60	90.41	92.64	92.07

FFAHSM-RDD 98.61 98.63 98.59 98.58 98.58 98.59

In the meantime, the existing methodologies, such as CNN+ SVM, Custom CNN model, CEViT-VDD, FFDL-DVD, DVD-CViT, ViXNet-DVIFD, and DD-DFSML, achieved worse performance under various measures.

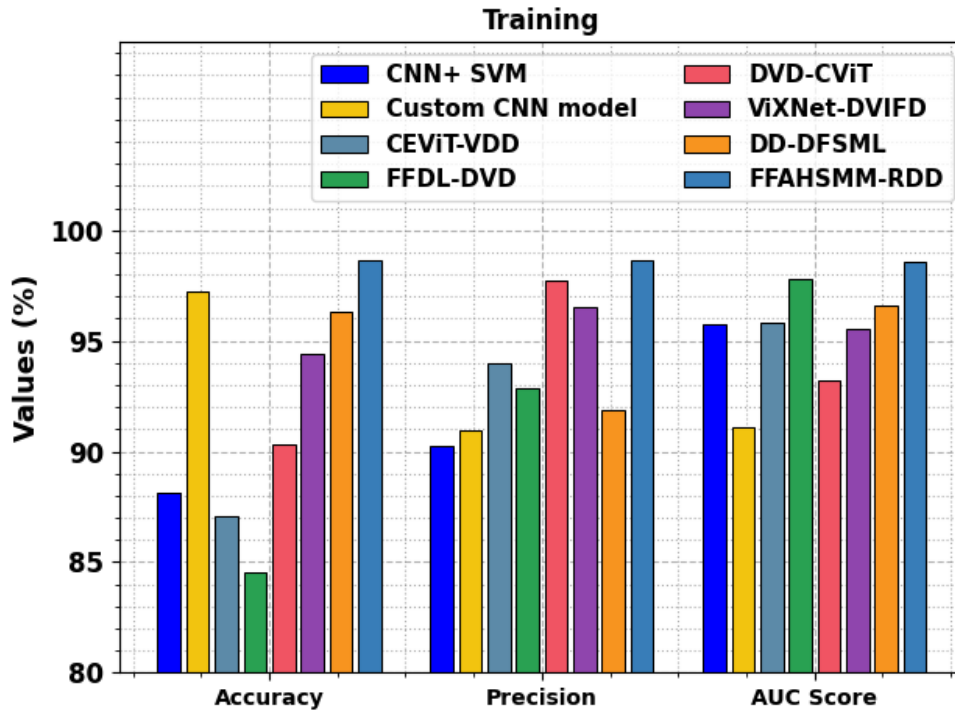


Fig. 14 Comparative analysis of the FFAHSM-RDD model under training

Fig. 15 describes the simulation outcome of the FFAHSM-RDD model with existing methodologies under testing. Under $accuracy$, the FFAHSM-RDD model got the highest $accuracy$ of 98.58%. In contrast, CNN+ SVM, Custom CNN model, CEViT-VDD, FFDL-DVD, DVD-CViT, ViXNet-DVIFD, and DD-DFSML methodologies have obtained a lower $accuracy$ of 95.95%, 91.36%, 93.88%, 92.07%, 94.43%, 96.41%, and 90.41%, respectively. Finally, on AUC_{Score} , the FFAHSM-RDD approach got a maximum AUC_{Score} of 98.59% while the CNN+ SVM, Custom CNN model, CEViT-VDD, FFDL-DVD, DVD-CViT, ViXNet-DVIFD, and DD-DFSML methodologies have attained a minimum AUC_{Score} of 89.14%, 91.36%, 98.38%, 90.12%, 94.93%, 91.80%, and 92.07%, respectively. These results indicate that approach demonstrates more consistent and superior overall discriminative capability across evaluation metrics.

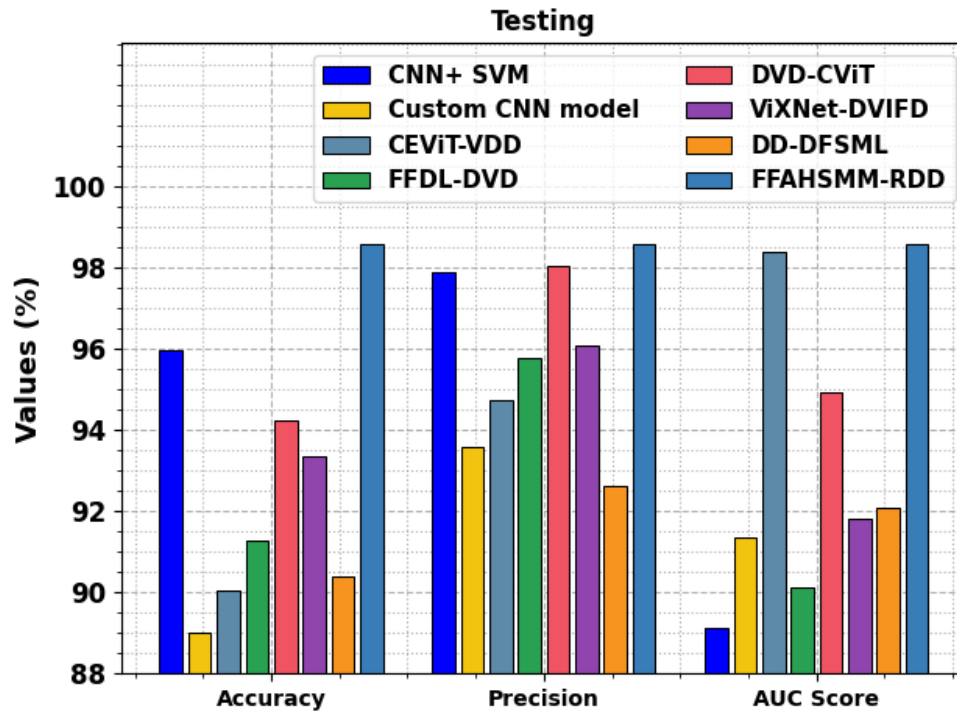


Fig. 15 Comparative analysis of the FFAHSMM-RDD model under testing

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4.3. Computational Time Analysis

Table 8 and Fig. 16 indicate the computational time (CT) evaluation of the FFAHSMM-RDD approach with existing methods. The FFAHSMM-RDD approach illustrated robust predictive capability with a CT of 2.45 sec, significantly lower than CNN plus SVM at 6.69 sec, Custom CNN model at 5.52 sec, CEViT-VDD at 6.22 sec, FFDL-DVD at 7.40 sec, DVD-CViT at 7.09 sec, ViXNet-DVIFD at 6.32 sec, and DD-DFSML at 6.73 sec. Thus, the model highlights superior computational efficiency while maintaining high classification performance.

Table 8 CT evaluation of the FFAHSMM-RDD approach with existing methods

Model	CT (sec)
CNN+ SVM	6.69
Custom CNN model	5.52
CEViT-VDD	6.22
FFDL-DVD	7.40
DVD-CViT	7.09

ViXNet-DVIFD	6.32
DD-DFSML	6.73
FFAHSM-RDD	2.45

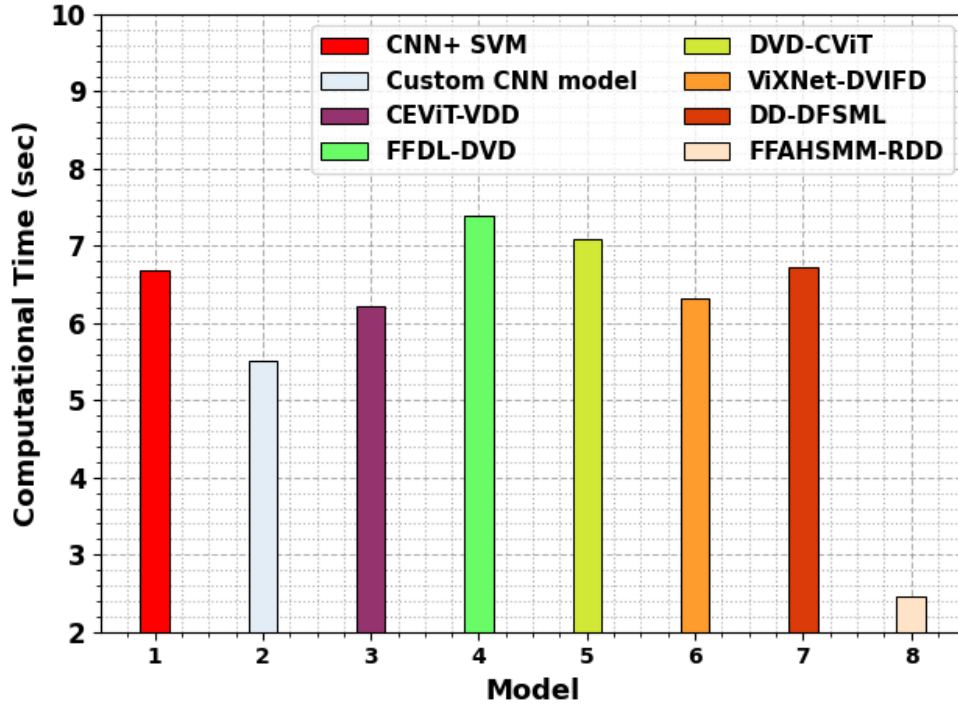


Fig. 16 CT evaluation of the FFAHSM-RDD approach with existing methods

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4.4. Ablation Study

Table 8 and Fig. 16 represent the ablation study of the FFAHSM-RDD model with existing methods. The DBN with fusion feature extraction and AHMM Sequence Modelling (FFAHSM-RDD) illustrated the efficiency of the model with an $accu_r$ of 98.61%, $preci_n$ of 98.63%, and AUC of 98.59%. Additionally, DBN+ResNet-50 achieved an $accu_r$ of 97.36%, $preci_n$ of 97.47%, and AUC of 97.33%. Furthermore, DBN+EfficientNet-B4 achieved an $accu_r$ of 97.65%, $preci_n$ of 97.68%, and AUC of 97.68%, respectively. Moreover, the DBN+MobileNetV3 achieved an $accu_r$ of 98.08%, $preci_n$ of 98.14%, and AUC of 98.07%, respectively. These results ensured that the proposed model has resulted in an effective deepfake detection process.

Table 8 Ablation study of FFAHSM-RDD model with existing methods

Methods	$Accur_y$	$Preci_n$	AUC_{Score}
DBN+ResNet-50 (Without Efficient-B4 and MobileNetV3)	97.36	97.47	97.33
DBN+EfficientNet-B4 (Without ResNet-50 and MobileNetV3)	97.65	97.68	97.68
DBN+MobileNetV3 (Without Efficient-B4 and ResNet-50)	98.08	98.14	98.07
FFAHSMM-RDD (DBN with fusion feature extraction and AHMM Sequence Modelling)	98.61	98.63	98.59

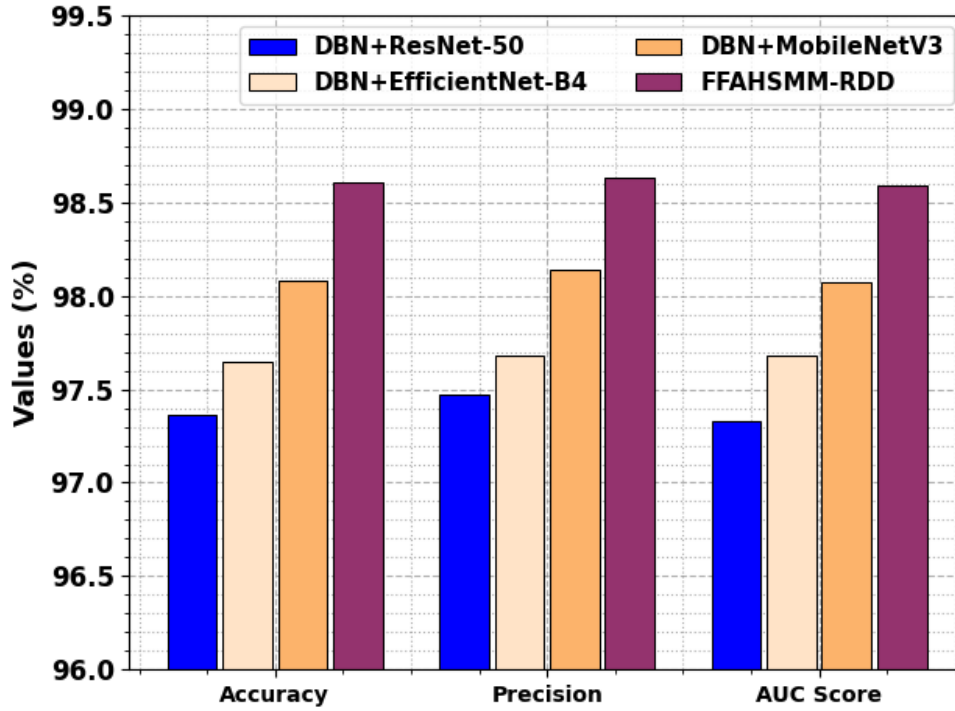


Fig. 16 Comparative analysis of ablation study under various models

5. Conclusion

In this article, the FFAHSMM-RDD model is presented. The FFAHSMM-RDD approach is designed to provide a reliable pathway for safeguarding multimedia communication and preventing deepfake-based threats in real-world applications. Primarily, the FFAHSMM-RDD approach uses data augmentation, image resizing, and normalization methods to enhance image quality and improve generality. For the feature extraction, a fusion of three methods, including ResNet-50, EfficientNet-B4, and MobileNetV3, is utilized for capturing diverse and discriminative features. Furthermore, the AHSMM approach is leveraged for

effectively modeling temporal dependencies and sequence dynamics in the data. In the deepfake classification process, a DBN is employed to effectively classify real and fake content. The performance validation of the FFAHSMM-RDD methodology demonstrated a superior value of 98.61% under the Celeb DF (v2) dataset. Limitations exist in low-quality videos and computationally heavy videos due to various CNN and DBN layers. Future work must explore lightweight, attention-based temporal, and self-supervised learning models for improving generalization and efficiency.

Data Availability Statement

The data that support the findings of this study are openly available in the Kaggle repository at <https://www.kaggle.com/datasets/reubensuju/celeb-df-v2>, reference number [26].

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